Stochastics - KVB

~> Discrete Probability Spaces

Random Experiment: => unpredictable outcome
=> repeatable

eg: (i) throw a die

- Sample space Ω set of all outcomes
  (i) $\Omega = \{1, 6\ 2, 3, 4, 5, 6\}$
- $\omega \in \Omega$ elementary Event (eg. 6)
- $A \subseteq \Omega$ Event (eg. $\{1, 2\}$)
- Power Set $\mathcal{P}(\Omega) = \{A : A \subseteq \Omega\}$
  eg: $\mathcal{P}(\{0,1\}) = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$
- A set M is at most countable
  <=> M is finite or the elements of M can be indexed by $\mathbb{N}$

~> Definition 2.1 (Discrete Probability Space)
A discrete probability space is a pair (Ω, P) of an at most countable set $\Omega \neq \emptyset$ and a function $P: \mathcal{P}(\Omega) \to [0,1]$ such that

(i) $P(\Omega) = 1$

(ii) for disjoint $A_1, A_2, \ldots \in \mathcal{P}(\Omega)$

$$P\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} P(A_k) \quad (\sigma\text{-additivity})$$

-> $P(\emptyset) = 0$

-> If $|\Omega| < \infty$, σ-additivity is equivalent to
$P(A \cup B) = P(A) + P(B)$ for $A, B \subseteq \Omega$ with $A \cap B = \emptyset$

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Example

- Throwing a die:

(Ω, P) with $\Omega = \{1,..,6\}$ and $P(A) = \frac{|A|}{6}$ for $A \subseteq \{1,..,6\}$

For example

$P(\text{'even'}) = P(\{2,4,6\}) = \frac{3}{6} = \frac{1}{2}$

(Ω, P) with $\Omega = \{1,..,6\}$ and

$$P(A) = \begin{cases} 1, & 6 \in A \\ 0, & 6 \notin A \end{cases} \quad \text{for } A \subseteq \{1,..,6\}$$

: another discrete probability space.

- Tossing a fair coin n times:

(Ω, P) with $\Omega = \{0,1\}^n$ and $P(A) = \frac{|A|}{2^n}$ for $A \subseteq \{0,1\}^n$

For example

$P(\text{'exactly one head'}) = P(\{(1,0,..,0),..,(0,..,0,1)\}) = \frac{n}{2^n}$

- Laplace Probability Space:

(Ω, P) with Ω a non-empty finite set and

$$P(A) = \frac{|A|}{|\Omega|} \quad \text{for } A \subseteq \Omega$$

- Tossing an unfair coin:

$p \in [0,1]$, (Ω, P) with $\Omega = \{0,1\}$ and P such that $P(\emptyset) = 0$; $P(\{0\}) = 1-p$, $P(\{1\}) = p$ and $P(\{1,0\}) = 1$.

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⇝ Lemma 2.8:

(i) For disjoint $A_1, \dots, A_n \subseteq \Omega$

$$P\left(\bigcup_{k=1}^{n} A_k\right) = \sum_{k=1}^{n} P(A_k)$$

(ii) For $A \subseteq \Omega$

$$P(A^c) = 1 - P(A) \quad (\Leftrightarrow) \quad P(A) + P(A^c) = 1$$

$A^c := \Omega \setminus A$

(iii) For $A \subseteq B \subseteq \Omega$,

$$P(A) \leq P(B)$$

(iv) For $A_1, \dots, A_n \subseteq \Omega$

$$P\left(\bigcup_{k=1}^{n} A_k\right) \leq \sum_{k=1}^{n} P(A_k)$$

⇝ Theorem 2.9: Inclusion-Exclusion-Principle

For $A_1, \dots, A_n \subseteq \Omega$, $n \geq 2$,

$$P\left(\bigcup_{k=1}^{n} A_k\right) = \sum_{k=1}^{n} (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n} P\left(\bigcap_{j=1}^{k} A_{i_j}\right)$$

e.g.:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$P(A \cup B \cup C \cup D) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(A \cap D) - P(B \cap C) - P(B \cap D) + P(C \cap D) + P(A \cap B \cap C) + P(A \cap B \cap D) + P(A \cap C \cap D) + P(B \cap C \cap D) - P(A \cap B \cap C \cap D)$$

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→ Elementary Combinatorics

Laplace Probability Space (Ω, P) with $0 < |\Omega| < \infty$ and

$$P(A) = \frac{|A|}{|\Omega|} \quad \text{for } A \subseteq \Omega$$

Urn Model: Goal: Number of choices to draw n of the M balls

→ replacement / no replacement
→ ordered, unordered samples

1) Ordered samples with replacement

$$\Omega_1 = \{(a_1, \dots, a_n) : a_1, \dots, a_n \in \{1, \dots, M\}\} = \{1, \dots, M\}^n$$

$$\Rightarrow |\Omega_1| = M^n$$

eg.) For $l, m \in \mathbb{N}$, $F_{l,m} = \{f : \{1, \dots, l\} \to \underbrace{\{1, \dots, m\}}_{U_m}\}$

$$|F_{l,m}| = m^l$$

2) Ordered Samples without replacement

$$\Omega_2 = \{(a_1, \dots, a_n) : a_1, \dots, a_n \in \{1, \dots, M\},\ a_i \neq a_j \text{ for } \underset{i \neq j}{i,j} \in \{1 \dots n\}\}$$

$$m! = m \cdot (m-1) \cdot \dots \cdot 2 \cdot 1 \quad \& \quad 0! = 1$$

$$|\Omega_2| = \frac{M!}{(M-n)!} \quad \text{for } n \leq M$$

eg.) For $l, m \in \mathbb{N}$ with $l \leq m$,

$$\bar{F}_{l,m} = \{g : \{1, \dots, l\} \to \{1, \dots, m\} : g(i) \neq g(j) \text{ for } \underset{i \neq j}{i,j} \in \{1 \dots l\}\}$$

$$\to |\bar{F}_{l,m}| = \frac{m!}{(m-l)!}$$

For $l = m$, $\bar{F}_{l,m}$ are the $\underbrace{\text{permutations of } \{1, \dots, l\}}_{\Pi_l}$

$$\to |\Pi_l| = l!$$

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$[a_1, \dots, a_n]$ unarded n-tuple, d.h. $[1,2] = [2,1]$

3) Unordered samples without replacement

$\Omega_3 = \{[a_1, \dots, a_n] : a_1, \dots, a_n \in \{1, \dots, M\}, a_i \neq a_j$ for $i,j \in \{1, \dots, n\}, i \neq j\}$

$= \{$ subsets of size n of $\{1, \dots, M\}\}$

$|\Omega_3| = \binom{M}{n} = \frac{M!}{(M-n)! \, n!}$, for $M \geq n$

Binomial Coefficient: $\binom{m}{k} = \begin{cases} \frac{m!}{k! (m-k)!}, & k \leq m \\ 0, & k > m \end{cases}$

→ Lemma 2.14:

For $m \in \mathbb{N}$ and $k \in \mathbb{N}_0$ such that $k \leq m$,

$$\binom{m}{k} = \binom{m-1}{k-1} + \binom{m-1}{k}$$

Pascals Triangle:

$\binom{m}{0} = 1$

```
            1
          1   1
        1   2   1
      1   3   3   1
```

→ Theorem 2.15:

For $n \in \mathbb{N}$ and $a, b \in \mathbb{R}$,

$$(a+b)^n = \sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i}$$

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4) Unordered samples with replacement

$$\Omega_4 := \{[a_1, \dots, a_n] : a_1, \dots, a_n \in \{1, \dots, M\}\}$$

$$|\Omega_4| = \binom{M+n-1}{n}$$

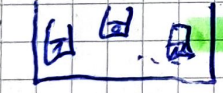
→ Balls into Bins

Goal: Number of choices to distribute the n balls into the M bins

(1) at most one ball / any number of balls per bin

(2) distinguishable / indistinguishable balls

Interpretation of ω as urn model:



→ draw n bins

→ k-th ball in the k-th bin

(1) $\hat{=}$ no replacement / replacement

(2) $\hat{=}$ ordered / unordered

Stochastics KVO - Ex 1

(1)

A 6-sided die is thrown twice

a) Specify a suitable probability space (Ω, P)

$\Omega = \{1, \dots, 6\}^2 \qquad P(A) = \frac{|A|}{|\Omega|} = \frac{|A|}{36}$

b) Give a set-theoretical description of the following events and calculate the probabilities

$A = \{$"the number 3 is thrown exactly once"$\}$

$= \{(a,b) \in \Omega^2 \mid a \neq b \wedge (a = 3 \vee b = 3)\}$

$= \{\underbrace{(3,1), \dots, (3,6)}_{5}, \underbrace{(1,3), \dots, (1,6)}_{5}\}$

$|A| = 10 \quad \rightsquigarrow P(A) = \frac{10}{36}$

$B = \{$"product is three in the second roll"$\}$

$= \{(a,b) : a \in \Omega, b \in \{2,3,5\}\}$

$|B| = 6 \cdot 3 = 18 \quad \rightsquigarrow P(B) = \frac{18}{36} = \frac{1}{2}$

$C = \{$"the eye sum is less than or equal to 11"$\}$

$= \{(a,b) : a,b \in \Omega;\ a+b \leq 11\}$

$C^c = \{$"eye sum is greater than 11"$\}$

$= \{(6,6)\}$

$P(C^c) = \frac{1}{36} \quad \rightsquigarrow P(C) = \frac{35}{36}$

Stochastik HVB – Ex 1

(2)

Let (Ω, P) be a discrete probability space and $A, B \subseteq \Omega$ with $P(A) = 0{,}7$ and $P(B) = 0{,}6$

a) e.g. $\Omega = \{1, \dots, 10\}$ $\quad P = \frac{|A|}{|\Omega|}$

$A = \{$"Draw a Card less than 8"$\} \rightarrow P(A) = 0{,}7$

$B = \{$"Draw a Card less than 7"$\} \rightarrow P(B) = 0{,}6$

b) In which case is

$A \cap B = A$ if $A \subseteq B$, but not possible since $A \subseteq B \Rightarrow P(A) \leq P(B)$ ↯

$A \cup B = A$ if $B \subseteq A$ → possible

c) Show that

$$0.3 \leq P(A \cap B) \leq 0{,}6$$

$P(A) = 0{,}7$, $P(A^c) = 0{,}3$, $P(B) = 0{,}6$

$B = B_1 \cup C$; $B_1 \subseteq A$, $C \subseteq A^c$

$C_{max} = A^c \Rightarrow 0{,}3 = P(A^c)$

$C_{min} = \emptyset \Rightarrow B_1 \subseteq A \Rightarrow P(A \cap B) = 0{,}6 = P(B)$

$$P(B) - P(A^c) \leq P(A \cap B) \leq P(B)$$

Better:

i) $(A \cap B) \subset B \Rightarrow P(A \subset B) \leq P(B) = 0{,}6$

ii) $P(A \cap B) = P(A) + P(B) - \underbrace{P(A \cup B)}_{\geq -1} \geq 0{,}7 + 0{,}6 - 1 = 0{,}3$ ✓

③

A number of $1, \dots, 1000\,000$ is randomly selected. Specify a suitable probability space and calculate the probability that the chosen number is divided by 2, 3 or 19

$$\Omega = \{1, \dots, 1000\,000\} \qquad P = \frac{|A|}{|\Omega|}$$

$A_1 = \{ a \in \Omega : a \bmod 2 = 0 \}$

$A_2 = \{ a \in \Omega : a \bmod 3 = 0 \}$

$A_{19} = \{ a \in \Omega : a \bmod 19 = 0 \}$

$$|A_1| = \frac{1\,000\,000}{2} \rightsquigarrow P(A_1) = \frac{1\,000\,000}{2 \cdot 1000000} = \frac{1}{2}$$

$$|A_2| = \left\lfloor \frac{1\,000\,000}{3} \right\rfloor \rightsquigarrow P(A_2) =$$

$$|A_3| = \left\lfloor \frac{1\,000\,000}{19} \right\rfloor \rightsquigarrow P(A_3) =$$

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_3) + P(A_2) - P(A_1 \cap A_3) - P(A_3 \cap A_2) - P(A_2 \cap A_1) + P(A_1 \cap A_2 \cap A_3) = 0{,}648$$

$$P(A_1 \cap A_2) = \frac{1\,000\,000}{2 \cdot 3} \cdot \frac{1}{1\,000\,000}$$

$$P(A_1 \cap A_2 \cap A_3) = \frac{1\,000\,000}{2 \cdot 3 \cdot 19} \cdot \frac{1}{1\,000\,000}$$

Schätze G1 – KVB

(4)

a) Determine the number of strictly increasing sequences $c_1 < c_2 < \dots < c_n$ with n elements and $c_i \in \{1, \dots, p\}$ for $i = 1, \dots, n$ and $n, p \in \mathbb{N}$

Unordered & no replacement

$$\rightsquigarrow |A| = \binom{p}{n}$$

b) Same for $c_0 \leq c_1 \leq \dots \leq c_n$

Unordered & replacement

$$|A| = \binom{p+n-1}{n}$$

Stochastik – KVB-G1

5)

How many five-digit numbers (00000 to 99999) exist, where

a) all digits are equal?

$\binom{10}{1} = 10$

b) exactly four digits are equal and the remaining one is different?

$\binom{10}{1} \cdot \binom{5}{4} \cdot \binom{9}{1} = 10 \cdot 5 \cdot 9 = 450$

c) exactly three digits are equal and the remaining two are different from each other?

$\binom{10}{1}\binom{9}{2}\binom{5}{3}\binom{2}{1}$

↳ 3 Digit ↳ 2 1 Digit → 3 Places → 1 Place

d) exactly three digits are equal and the remaining two are also equal?

$\binom{10}{1}\binom{9}{1}\binom{5}{3}$

e) exactly two digits are equal and the remaining three are different?

$\binom{10}{1}\binom{9}{3}\binom{5}{2}$ [illegible] $\binom{3}{1}\binom{2}{1}$

↳ 2 Diff ↳ 3 1 Digit → 2 Place [illegible]

f) two pairs of equal digits occur which are different but each like 1 Digit?

$\binom{10}{2}\binom{8}{1}\binom{5}{2}\binom{3}{2}$

g) 5 diff Digits?

$\binom{10}{5}\binom{5}{1}\binom{4}{1}\binom{3}{1}\binom{2}{1}$

Stochastics - K.V.B

→ Discrete probability distributions

For a discrete probability (Ω, P) and $A \subseteq \Omega$,

$$P(A) = P\left(\bigcup_{\omega \in A} \{\omega\}\right) = \sum_{\omega \in A} P(A)$$

→ Theorem 2.19

Let $\Omega \neq \emptyset$ be an at most countable set and let $(p_\omega)_{\omega \in \Omega}$ be a non negative real numbers such that

$$\sum_{\omega \in \Omega} p_\omega = 1$$

Then, (Ω, P) with

$$P(A) = \sum_{\omega \in A} p_\omega \quad \text{for } A \subseteq \Omega$$

$\hookrightarrow = P(\{\omega\})$

is a discrete probability space

→ Laplace Distribution:

For $\Omega \neq \emptyset$ finite set

$$p_\omega = \frac{1}{|\Omega|} \quad \text{, for } \omega \in \Omega$$

so that $p_\omega \geq 0$ for $\omega \in \Omega$ and

$$\sum_{\omega \in \Omega} p_\omega = \sum_{\omega \in \Omega} \frac{1}{|\Omega|} = \frac{|\Omega|}{|\Omega|} = 1$$

By Theorem 2.19 (Ω, P) with

$$P(A) = \sum_{\omega \in A} p_\omega = \sum_{\omega \in A} \frac{1}{|\Omega|} = \frac{|A|}{|\Omega|}$$

is a discrete probability space, namely Laplace probability space.

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Dirac Distribution

For $\Omega \neq \emptyset$ at most countable and a fixed $\omega_0 \in \Omega$

Let

$$p_\omega = \begin{cases} 1, & \omega = \omega_0 \\ 0, & \omega \neq \omega_0 \end{cases} \quad \text{for } \omega \in \Omega$$

Since $p_\omega \geq 0$ for $\omega \in \Omega$ and $\sum_{\omega \in \Omega} p_\omega = 1$, by Theorem 2.14

(Ω, P) with

$$P(A) = \sum_{\omega \in A} p_\omega = \begin{cases} 1, & \omega_0 \in A \\ 0, & \omega_0 \notin A \end{cases} \quad \text{for } A \subseteq \Omega$$

is a discrete probability space.

Theorem 2.22

For $m \in \mathbb{N}$ and discrete probability spaces $(\Omega_1, P_1), \dots, (\Omega_m, P_m)$

define $\Omega = \bigtimes_{i=1}^{m} \Omega_i$ and

$$P(A) = \sum_{(\omega_1, \dots, \omega_m) \in A} \underbrace{\prod_{i=1}^{m} P_i(\{\omega_i\})}_{= P((\omega_1, \dots, \omega_m))}, \quad \text{for } A \subseteq \Omega$$

Then, (Ω, P) is a discrete probability space

product probability space

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→ Rolling a die twice

Let $\Omega_D = \{1,\dots,6\}$ and $P_D(A) = \frac{|A|}{6}$ for $A \subseteq \Omega_D$.
The product probability space (Ω, P) of (Ω_D, P_D) with itself is given by

$$\Omega = \Omega_D \times \Omega_D = \Omega_D^2 = \{1,\dots,6\}^2$$

and

$$P(A) = \sum_{(\omega_1,\omega_2)\in A} P_D(\{\omega_1\})\,P_D(\{\omega_2\}) = \sum_{(\omega_1,\omega_2)\in A} \frac{1}{36}$$

Since $P(A) = \frac{|A|}{36}$, P is the Laplace distribution

→ Tossing an unfair coin n times

Let (Ω_C, P_C) be such that $\Omega_C = \{0,1\}$ and $P_C(\{0\}) = 1-p$ and $P_C(\{1\}) = p$ with $p \in [0,1]$. For $n \in \mathbb{N}$ the n-fold product probability space (Ω, P) of (Ω_C, P_C) is given by

$$\Omega = \{0,1\}^n$$

and

$$P(A) = \sum_{(\omega_1,\dots,\omega_n)\in A} \prod_{i=1}^{n} P_C(\{\omega_i\}) = \sum_{(\omega_1,\dots,\omega_n)\in A} \prod_{i=1}^{n} p^{\omega_i}(1-p)^{1-\omega_i}$$

The n-fold repetition of a random experiment with the outcomes success/1 & failure/0 is described by (Ω, P).
$k \in \{0,1,\dots,n\}$ the probability of k successes and $n-k$ failures in a fixed order is

$$p^k (1-p)^{n-k}$$

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→ Binomial Distribution

For $n \in \mathbb{N}$ and $p \in [0,1]$ define

$$p_k = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for } k \in \{0,\dots,n\}$$ // exactly k successes

Since $p_k \geq 0$ for $k \in \{0,\dots,n\}$ and

$$\sum_{k=0}^{n} p_k = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p+1-p)^n = 1$$

Theorem 2.19 yields that (Ω, P) with $\Omega = \{0,\dots,n\}$ and

$$P(A) = \sum_{k \in A} \binom{n}{k} p^k (1-p)^{n-k} \quad \text{, for } A \subseteq \Omega$$

is a discrete probability space

→ Poisson Distribution

For $\lambda \in (0,\infty)$ define

$$p_k = \frac{\lambda^k}{k!} e^{-\lambda} \quad \text{for } k \in \mathbb{N}_0$$

Since $p_k \geq 0$ for $k \in \mathbb{N}$ and

$$\sum_{k=0}^{\infty} p_k = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1$$

(Exponential Series)

by Theorem 2.19,

$$P(A) = \sum_{k \in A} \frac{\lambda^k}{k!} e^{-\lambda} \quad \text{for } A \subseteq \mathbb{N}_0$$

is a probability distribution on $\mathbb{N}_0$

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↳ Geometric distribution

For $p \in (0,1)$ ↳ number of failures till first success

$$P(A) = \sum_{k \in A} p^k (1-p) \quad \text{for } A \subseteq \mathbb{N}_0$$

is a probability distribution on $\mathbb{N}_0$. This follows from Theorem 2.19 since $p_k \geq 0$ $k \in \mathbb{N}_0$ and

$$\sum_{k=0}^{\infty} p_k = \sum_{k=0}^{\infty} p^k (1-p) = (1-p) \sum_{k=0}^{\infty} p^k = (1-p)\frac{1}{1-p} = 1$$

Geometric series $\sum_{k=0}^{\infty} p^k = \frac{1}{1-p}$ for $p < 1$

Stochastics - KVB

(1)

Let A, B be events on some probability space. Show that

a) $P(B \setminus A) = P(B) - P(A)$ for $A \subseteq B$

$B = \underbrace{B \setminus A \cup A}_{\text{disjoint}} \Rightarrow P(B) = P(B \setminus A) + P(A)$

$\Leftrightarrow P(B \setminus A) = P(B) - P(A)$ ▨

b) $P(A \cap B) = P(A) - P(A \cap B^c)$

$A = \underbrace{A \cap B + A \cap B^c}_{\text{disjoint}} \Rightarrow P(A) = P(A \cap B) + P(A \cap B^c)$

$\Leftrightarrow P(A \cap B) = P(A) - P(A \cap B^c)$ ▨

Stochastics KIB Ex 2

(2)

An urn contains n balls, which are numbered from 1 to n. One chooses a ball at random and puts it back into the urn. This is repeated r times. Specify a suitable probability space & determine the probability that the largest selected number is equal to some $m \in \{1, \dots, n\}$

Hint: Consider also the event that the maximum among the r draws is less than or equal to some $k \in \{1, \dots, n\}$

$$\Omega = \{1, \dots, n\}^r$$

$$P(A) = \frac{|A|}{n^r}$$

$B = \{$"max among the r draws is less than or equal to some $k \in \{1, \dots, n\}$"$\}$

$$A_m = \{(\omega_1, \dots, \omega_r) \in \Omega,\ \omega_i \leq m,\ \exists\, i : \omega_i = m\}$$

$$B_k = \{(\omega_1, \dots, \omega_r) : \omega_i \in \{1, \dots, k\}\} = \{1, \dots, k\}^r$$

$A_m = B_m \setminus B_{m-1}$ $\qquad$ $B_{m-1} \subseteq B_m$

$$P(A_m) = P(B_m \setminus B_{m-1}) = P(B_m) - P(B_{m-1})$$

$$= \frac{|B_m|}{n^r} - \frac{|B_{m-1}|}{n^r} = \frac{m^r}{n^r} - \frac{(m-1)^r}{n^r} = \frac{m^r - (m-1)^r}{n^r}$$

Stochastics - KW 3 - Ex 2

③

There are seven passengers on a bus and there are four more stops to the final stop. Specify a suitable probability space and calculate the probability that at least one person gets off at each stop.

$\Omega = \{1,\dots,4\}^7$

$P = \frac{|A|}{4^7}$

$A = \{$"Each Stop one passenger gores off"$\}$

$|A| = \underline{4}\cdot\underline{3}\cdot\underline{2}\cdot\underline{1}\cdot[\ \ \ \ ] \hookrightarrow 4^3\cdot 7!$

$B_i = \{$"At least one at stop i"$\}$

$= \{(\omega_1,\dots,\omega_7) \in \Omega : i \in \{\omega_1,\dots,\omega_7\}\}$

$\Leftrightarrow A = B_1 \cap B_2 \cap B_3 \cap B_4$

Demorgan $[(A\cap B)^c = A^c \cup B^c]$

$P(B_1 \cap B_2 \cap B_3 \cap B_4) = 1 - P((B_1 \cap B_2 \cap B_3 \cap B_4)^c)$

$= 1 - P(B_1^c \cup B_2^c \cup B_3^c \cup B_4^c)$

$B_1^c = \{2,3,4\}^7 \rightsquigarrow |B_1^c| = 3^7$

Stochastik - KVB - G2

③

$$P(B_1 \cap B_2 \cap B_3 \cap B_4) = 1 - P(B_1^c \cup B_2^c \cup B_3^c \cup B_4^c)$$

$$= 1 - \big(P(B_1^c) + P(B_2^c) + P(B_3^c) + P(B_4^c) - P(B_1^c \cap B_2^c)$$
$$- P(B_1^c \cap B_3^c) - P(B_1^c \cap B_4^c) - P(B_2^c \cap B_3^c) - P(B_2^c \cap B_4^c)$$
$$- P(B_3^c \cap B_4^c) + P(B_1^c \cap B_2^c \cap B_3^c) + P(B_1^c \cap B_3^c \cap B_4^c)$$
$$+ P(B_1^c \cap B_2^c \cap B_4^c) + P(B_2^c \cap B_3^c \cap B_4^c) - P(B_1^c \cap B_2^c \cap B_3^c \cap B_4^c)$$

$$= 0{,}512$$

↳ $P(A_i^c) = P(A_1^c) = \frac{3^7}{4^7}$

↳ $P(A_i^c \cap A_j^c) = P(A_1^c \cap A_2^c) = \frac{2^7}{4^7}$

↳ $P(A_a^c \cap A_b^c \cap A_c^c) = P(A_1^c \cap A_3^c \cap A_2^c) = \frac{1^7}{4^7}$

↳ $P(A_1^c \cap A_2^c \cap A_3^c \cap A_4^c) = 0$

Stochastik – KVB – Ex 2

(4)

Let $p \in (0,1)$ and $m \in \mathbb{N}_0$. Show that

$$P(A) = \sum_{k \in A} (1-p)^{k-m} p$$

for $A \subseteq \{m, m+1, \dots\}$ defines a discrete probability distribution on $\{m, m+1, \dots\}$

$$\rightsquigarrow \sum_{k \in A} (1-p)^{k-m} p = p \sum_{k \in \{m, m+1, \dots\}} (1-p)^{k-m} = p \sum_{s=0}^{\infty} (1-p)^s$$

$(1-p) \leq 1$

$$= p \frac{1}{1-(1-p)} = p \frac{1}{p} = 1$$

$p_k = (1-p)^{k-m} p$

$\rightsquigarrow p_k \geq 0 \quad \Rightarrow$ Lg 2.19.

$$P(A) = \sum_{k \in A} (1-p)^{k-m} p$$

is a probability distribution on $\{m, m+1, \dots\}$

Stochastik - KVB - Ex 2

(5)

A coin is tossed 1000 000 times

a) Use R to simulate these coin tosses by generating the integers 0 & 1 randomly with the same probability

$$P(A) = \sum_{k \in A} \binom{1}{k} 0{,}5^{k} 0{,}5^{1-k}$$

↳ Binomial Distribution

$\Omega = \{0,1\}^{1000000}$

```
# coin tossing
toss = rbinom(1000000, 1, 0.5)
```

b) Calculate frequency for Head

```
head = numeric(1000)  // like vector
for (i in seq(100, 1000000, by = 100)) {
    head[i/100] = mean(toss[1:i])
}
```

c) Plot

```
plot(seq(100, 1000000, by = 100), head, 'l', main "...",
     xlab = '...', ylab = '...')
```